Active Maintenance Report: coppock_green_2016

Coppock and Green (2016, AJPS): Is Voting Habit Forming?

Table of contents

Drafted by Claude Opus 5 under the supervision of Alex Coppock.

This repository holds the actively maintained replication code for Coppock and Green (2016), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1111/ajps.12210
Replication archive	10.7910/DVN/ALZVAW
Errata	`coppock_green_2016_errata.pdf`

The data are not redistributed here. The deposit is 77 MB across 24 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This README is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_green_2016.Rproj`, and run:

```r
source("run_all.R")
```

That fetches the deposit, verifies its 24 files, produces every table and figure into `maintained/output/`, rebuilds the ground truth and runs the coverage gate. Required packages: tidyverse, estimatr, metafor, sandwich, modelsummary, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.**

Two scripts in `ground_truth/` are not part of `run_all.R`, because they read things this repository does not carry. `run_archive.R` runs the deposit's own ten scripts in a scratch copy and `extract_archive_values.R` parses what they printed; `extract_published_values.R` reads the published article and its supporting information. Both write committed CSVs, and both take the directory they need from an environment variable.

1 Paper overview

Citation: Coppock, A. and Green, D. P. (2016). “Is Voting Habit Forming? New Evidence from Experiments and Regression Discontinuities.” *American Journal of Political Science*, 60(4), 1044–1062. DOI: 10.1111/ajps.12210

Summary: This paper tests whether voting is habit forming using two research designs. The experimental design exploits three randomized GOTV field experiments (ETOV 2006, ETOV 2007, and SMG 2009) in which randomly assigned treatment arms induced upstream voting; the downstream effect on voting in subsequent elections identifies the Complier Average Causal Effect (CACE) of upstream voting. The regression discontinuity design exploits the sharp 18th birthday eligibility cutoff: individuals just old enough to vote in an upstream election can be compared to those just too young, identifying the CACE from a different source of variation. Both designs produce positive downstream estimates, with same-type election pairs (presidential-on-presidential, midterm-on-midterm) generating the largest and most persistent effects.

---

2 Summary

Two questions, answered before the detail.

2.1 Does the deposited archive run?

Nine of the ten analysis scripts do, once two packages are installed. `AER` supplies `ivreg` and `rmeta` supplies `meta.summaries`; both remain on CRAN and install without incident, rechecked 9 August 2026. Nothing else stops those nine: no hardcoded paths to a machine that no longer exists, no functions called before they are defined.

The tenth does not run. `CG Habit RD Figure A2.R` indexes one position past the end of its own loop on the last upstream election, and the resulting subscript error is not inside the `try()` the loop wraps around the estimator. The script stops before it assembles the frame it plots, so neither Figure A2 nor the 60-day primary-election analysis printed beneath it is ever reached. The maintained rewrite produces both.

One script also cannot find its data. `CG Habit ME Table A2 and Figure A1.R` reads `Measurement Error/interstatemovers.txt`; the deposit ships `interstatemovers.txt` at the top level and no such directory. `ground_truth/run_archive.R` puts a copy where the code looks, and says so.

Two further facts about the deposit are worth recording. Running it writes nine files into its working directory: four `.tex` tables, four household tables and an `Rplots.pdf`, which is why it is never run inside `original/`. Four of its tables are built as `xtable` objects whose print calls are commented out, so those scripts compute Table 6 and Tables A6 and A7 and display nothing at all.

2.2 Does the maintained rewrite reproduce the paper?

Almost exactly. The ground truth carries 1,457 rows: 1,329 published table cells compared one at a time, and 128 claims the article states in prose. 1,397 reproduce at the precision the page prints, 18 do not, and 23 of the 27 claims with a computed truth value hold.

The 18 that do not divide into three groups: eleven standard errors that differ from the published ones in the fifth decimal, where two packages define the same nominal estimator differently; three cells of one appendix table row that the deposit fills from a typed constant its own analysis does not produce; and four prose quantities that miscount or misstate something the article prints correctly. The four claims that do not hold are as many more sentences of the same kind. All eight of the article's own errors are set out in the errata, along with one more that no ground truth row reaches; the eleven standard errors are not errors in the article and belong to the software rather than to either analysis.

---

3 Original archive reproducibility

Table 1: Every deposited script, run in a scratch copy of the archive rather than in place.

Deposited script	Status on a current R installation
CG Habit DE Table 1.R	ok
CG Habit DE Table 2.R	ok
CG Habit DE Table 3.R	ok
CG Habit DE Table A9.R	ok
CG Habit ME Table A2 and Figure A1.R	ok
CG Habit RD Tables 4 and 5.R	ok
CG Habit RD Table 6.R	ok
CG Habit RD Table 7.R	ok
CG Habit RD Tables A6 and A7.R	ok
CG Habit RD Figure A2.R	Error: subscript out of bounds

The deposit ships no intermediate objects: its 24 files are ten R scripts, a helper file, five data files, four codebooks, a README and one text file of migration counts. There is therefore nothing to strip, and the usual test of running the archive without its own saved intermediates has nothing to remove. That is asserted from the manifest rather than assumed.

Category 2 in this program's vocabulary, but the category records whether code executes, not whether the numbers reproduce. Those are separate questions and the next section answers the second.

---

4 The extraction and the two instruments

Two files stand between the published pages and the code, and they are deliberately independent of one another.

`ground_truth/published_claims.csv` is the extraction: every numeric token in the article and its supporting information, read line by line rather than searched for, classified by hand into the five claim types this program uses, and carrying the precision at which the page prints each one. It has 210 rows.

Table 2: The extraction, by claim type and whether the second instrument must print it.

Claim type	No block	Block required
definitional	24	5
descriptive	0	35
pipeline	0	88
structural	21	0
transcribed	37	0

The coverage boundary. Every number in the article's body, its footnotes, its table notes and its supporting information is in the extraction, including numbers spelled as words. Five classes are excluded, and nothing else is: bibliographic years and page numbers in citations, and the reference list entire; years used as an election or study label, where "the August 2006 primary" names a period rather than asserting a quantity; equation, section, table and figure numbers, and footnote markers; the affiliation, address and IRB identifiers on the title page; and the journal's own volume, issue, page-range and DOI line. Sentences that define what a column of a table contains are in, whether or not they carry a digit, and three of the corrections below descend from such a sentence.

`maintained/in_text_claims.R` is the second instrument. It recomputes every one of the 128 claims the extraction marks as needing a block, reading only the pipeline's output and reaching each quantity by a path of its own: where the ground truth selects a row by the label the published table prints, the claims file selects it by the treatment arm the deposit names. It prints one line per claim, and `ground_truth/build_ground_truth.R` runs it as a program, counts what it printed, and compares each printed value against the ground truth's own. Where the two disagree, one of them is wrong; three such disagreements surfaced while this was being built, and all three were errors in the checking code rather than in the pipeline.

What the gate asserts, in order. The checks that depend only on the extraction run first, so a wrong precision trips its own check rather than a value comparison downstream: no duplicate claim ids, a known claim type and comparison mode on every row, a block required for every `pipeline` and `descriptive` claim, and a round trip proving each stored `value_paper` renders back to itself at its own recorded precision. Then the locus rule in three states, then the extraction reconciled against the ground truth's transcription of the same pages, then the float inventory, then the second instrument. The gate was tested by breaking it three ways: deleting a block fails the count, corrupting a computed value fails the cross-instrument comparison, and corrupting a `digits` entry fails the round trip before anything consumes it.

---

5 Number-by-number comparison

`value_paper` comes only from the published pages. The 1,329 table cells were read off the two PDFs by `ground_truth/extract_published_values.R`, positionally where a table has holes in it and in reading order where it does not, and spot-checked against rendered pages. `value_script` is what the deposit's own scripts printed. `value_rewrite` is read out of `maintained/output/`. No published number is an input to any computation in `maintained/`.

Table 3: Ground truth verdicts. A row has no verdict where the quantity is not one the deposit prints, where the claim is a hedge, or where its verdict is a truth value rather than a number.

Comparison	Agree	Disagree	No verdict
The deposit against the published pages	1314	2	141
The rewrite against the published pages	1397	18	42

5.1 Float coverage

Table 4: Every published float, the numbers it prints, and how many of them this repository compares.

Float	Numbers printed	Covered	Rewrite reproduces	Deposit reproduces
Table 1	123	123	123	116
Table 2	92	92	92	86
Table 3	78	78	78	78
Table 4	238	238	237	238

Table 5	198	198	197	198
Table 6	192	192	190	192
Table 7	53	53	52	51
Figure A1	0	0	–	–
Table A1	72	0	–	–
Table A2	75	75	72	75
Table A3	72	0	–	–
Table A4	264	0	–	–
Table A5	171	0	–	–
Table A6	128	128	126	128
Table A7	128	128	124	128
Figure A2	0	0	–	–
Table A8	40	0	–	–
Table A9	24	24	24	24

The article and its supporting information print 1,948 numbers across 18 floats, of which this repository compares 1,329, or 68 per cent. Six floats have no coverage, and each has a reason that was tested rather than assumed:

- **Tables A1 and A3** are typologies of measurement-error cases, enumerating twelve types of residentially mobile voter. Nothing in the deposit produces them and nothing could: they are a taxonomy, not an analysis output.
- **Tables A4 and A5** give statewide official turnout and statewide counts of votes recorded on each voter file. Official turnout comes from an external website. The counts cannot be recomputed from the deposit, which aggregates each voter file to votes cast per birthdate cohort within a narrow band around the eligibility cutoff: summing Illinois's 2012 column of the deposited file gives 1,682,532 against the 5,175,513 the table prints, because the deposited file is not the voter file.
- **Table A8** has neither code nor data in the deposit. Its ANES file is confidential, which the deposit's own README states.
- **Figure A1** draws a continuous bias surface and prints no estimate on its face, so there is nothing discrete to count. The rewrite commits the surface at the ten complier counts the figure's own legend labels, and the three quantities the page does state (the assumed true CACE, the range of the horizontal axis and the number of legend labels) are claims in the extraction with blocks of their own.
- **Figure A2** prints no numbers either. What it states is a count of plotted estimates, which is a claim with a block.

5.2 Where the numbers disagree

Table 5: Every row where the deposit or the rewrite disagrees with the published page, or where a claim does not hold.

Location	Quantity	Paper	Rewrite	Locus

Table 4	Iowa, 2004-06 (Presidential on Midterm), se	0.049	0.048498	environment
Table 5	Kentucky, 2008-12 (Presidential on Presidential), se	0.022	0.021496	environment
Table 6	Kentucky, 2008-12 (lower), se	0.022	0.021496	environment
Table 6	Meta-Analysis, 2002-12 (lower), est	0.271	0.2715	environment
Table 7	Presidential upstream, (5), se	0.0207	0.020649	environment
Table 7	State fixed effects, (4), est	Yes	1	environment
Table 7	State fixed effects, (5), est	Yes	1	environment
Table A2	IA, CACE 08-12, est	0.086	0.081	archive
Table A2	IA, Bias Estimate, est	-0.008	-0.0071769	archive
Table A2	IA, Corrected CACE, est	0.094	0.088177	archive
Table A6	Second-order Polynomial, 90 Days (No additional controls), est	-0.028	-0.027383	environment
Table A6	Third-order Polynomial, 635 Days (Controls for lagged vote totals), se	0.009	0.0084919	environment
Table A7	Third-order Polynomial, 90 Days (No additional controls), est	-0.019	-0.019576	environment
Table A7	Third-order Polynomial, 545 Days (No additional controls), est	0.101	0.10049	environment
Table A7	Second-order Polynomial, 90 Days (Controls for lagged vote totals), est	-0.006	-0.0066084	environment
Table A7	Third-order Polynomial, 90 Days (Controls for lagged vote totals), est	0.002	0.0011334	environment
Table 4	Positive midterm-on-presidential estimates	50	51	paper_internal
Table 4	Midterm-on-presidential state estimates	54	55	paper_internal
Table 5	Four of the five strongest 2008-12 estimates in nonbattleground states	–	FALSE	paper_internal
Table 7	Battleground coefficient falls by a factor of three between columns 2 and 3	3	FALSE	paper_internal
Table A2	States in Table A2	16	15	paper_internal
Table A2	Table A2's CACE column holds the 2008 on 2012 estimate of Tables 5 and 6	–	FALSE	archive
SI section 3	Those pairs are the ones Table 6 of the main text reports	–	FALSE	paper_internal
Figure A2	Its robust standard error	0.0010	0.0099495	paper_internal

Eleven standard errors, locus `environment`. The deposit estimates each discontinuity with `AER::ivreg` and takes its standard error from `sandwich::vcovHC`; the rewrite uses `estimatr::iv_robust(se_type = "HC3")`. The point estimates agree to ten decimal places everywhere. The standard errors differ by about a part in a thousand, because the two packages define the HC3 leverage adjustment for two-stage least squares differently, and that is enough to move a third decimal on the eleven cells that already sat within a whisker of a rounding boundary. Every one of the eleven is a cell the deposit reproduces exactly, so the difference is between two toolchains rather than in either analysis.

Two cells of Table 7, locus `environment`. The published table records state fixed effects in columns 4 and 5, which is what those models fit. The deposit's `stargazer` call prints "No" in all five columns under a current `stargazer`, so the deposit no longer reproduces its own table's fixed-effects row. The rewrite's `modelsummary` table gets it right.

Three cells of Table A2, locus `archive`. See errata entry 2.

Four sentences, locus `paper_internal`, and one further one. See errata entries 1 and 3 through 7.

6 Errata

Nine corrections, in `coppock_green_2016_errata.pdf` at the root of this repository, generated from the pipeline with every corrected value computed at render time. **None of them changes a conclusion of the paper.** In the note's order and with its numbering:

1. The standard error of the primary-on-primary estimate (published, supporting information)
2. Appendix Table A2's Iowa row (published, appendix Table A2)
3. The count of midterm-on-presidential estimates (published p. 1053)
4. Battleground status among the strongest presidential-on-presidential estimates (published p. 1059)
5. The change in the battleground coefficient between columns 2 and 3 of Table 7 (published, main text)
6. The number of states in Table A2 (published, appendix)
7. The cross-reference for the general-on-general election pairs (published, appendix)
8. The first panel of appendix Table A4 (published, appendix Table A4)
9. Atkinson and Fowler (2014) prints the last page of its range without the first (published p. 1061)

The last one is in the reference list and no ground truth row reaches it: every printed entry was sent whole to Crossref and the authoritative record checked back into it. The audit flagged two entries for a page range, and only this one is real. Angrist, Imbens and Rubin (1996) prints "91(434): 444–55", which is the abbreviation the reference style uses throughout; Atkinson and Fowler (2014) prints "44(1): 59", where the range is 41 to 59 and the first number is missing.

6.1 Corrections the rewrite made to the deposited code

The rewrite corrects two clear coding errors in the deposit. Neither is an analytical decision.

`CG Habit ME Table A2 and Figure A1.R` types its fifteen CACE estimates in as constants. The rewrite reads them from the regression discontinuity output instead, which is where fourteen of them came from; the fifteenth is errata entry 2.

`CG Habit RD Figure A2.R` stops at a subscript error before producing anything. The rewrite's grid is built by `crossing` rather than by an index loop, so the error has no analogue, and the 60-day primary-election analysis the deposited script never reaches runs and is written to `maintained/output/text_like_elections.csv`.

One defect in an earlier version of this rewrite is worth recording because it is the kind a value comparison cannot see. Its migration matrix was built with `pivot_wider`, which orders new columns by first appearance, and the code then renamed those columns to the sorted row order. The column sums and the diagonal were consequently taken from the wrong states, and every figure in Table A2 was wrong by two orders of magnitude. Sorting the columns and asserting that the two margins carry the same states in the same order fixes it, and the assertion is what would catch a recurrence.

---

7 Maintained rewrite

The maintained rewrite translates the deposit's ten scripts into eleven, following the project style guide, plus the second instrument. The rewrite is a translation: all analytical decisions, estimators and sample restrictions are preserved.

7.1 Architecture

Downstream experiments. The deposit repeats an `ivreg` plus `cl()` call for each treatment arm by downstream election cell, roughly thirty to forty calls per script. The rewrite pivots each dataset long on the downstream election columns, then maps `iv_robust(se_type = "stata")` within each cell via `nest_by(election)`, with `pmap_dfr` over the treatment arms.

Regression discontinuity. The deposit uses nested `for` loops with `try()` wrappers. The rewrite builds a crossing of state by year-pair and calls `pmap` with a `run_rd_iv()` helper wrapping `iv_robust(se_type = "HC3")`.

Meta-analysis. The deposit uses `rmeta::meta.summaries`, which defaults to fixed effects. The rewrite uses `metafor::rma(method = "FE")` for the published estimates and additionally computes `rma(method = "REML")`, reported alongside rather than instead.

Table 6: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
‘rm(list = ls())‘	(omitted)
‘setwd(\”\”)‘	‘here::here()‘
‘library(AER)‘ + ‘ivreg()‘ + ‘cl()‘	‘estimatr::iv_robust(se_type = ”stata”)‘ for clustered DE
‘coeftest(fit, vcovHC)[2,2]‘	‘estimatr::iv_robust(se_type = ”HC3”)‘ for RD
‘library(rmeta)‘ + ‘meta.summaries()‘	‘metafor::rma(method = ”FE”)‘ + ‘rma(method = ”REML”)‘
‘library(xtable)‘ + ‘print.xtable()‘	‘write_csv()‘ to ‘output/‘
‘library(stargazer)‘ + ‘sink()‘	‘modelsummary(output = ...)‘
‘plyr::adply(.margins = c(1,2,3))‘	‘pmap_dfr()‘ over ‘crossing()‘ grid
‘library(beepr)‘ + ‘beep()‘	(omitted)
‘system(”say Just finished!”)‘	(omitted)

‘library(reshape2)‘ + ‘dcast()‘ + ‘melt()‘	‘pivot_wider()‘ + ‘pivot_longer()‘
‘%>%‘ pipes	‘|>‘ native pipe

8 Figure verification

The article prints no figures; the supporting information prints two, and each figure script writes a CSV of what it draws.

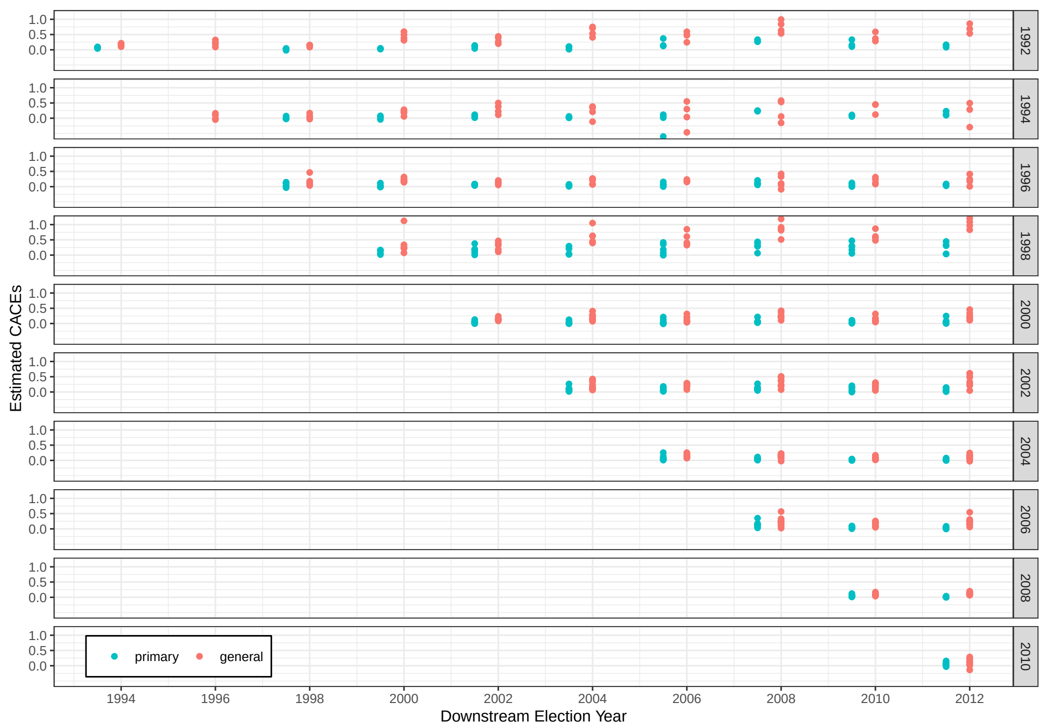

Figure A2 plots one point per upstream-downstream-state combination for the ten upstream general elections from 1992 to 2010, coloured by whether the downstream election is a primary or a general. The rewrite’s version was laid beside the published page: same ten panels, same axis breaks, same legend placement, same sawtooth. `maintained/output/figure_a2_data.csv` holds every plotted estimate and its standard error.

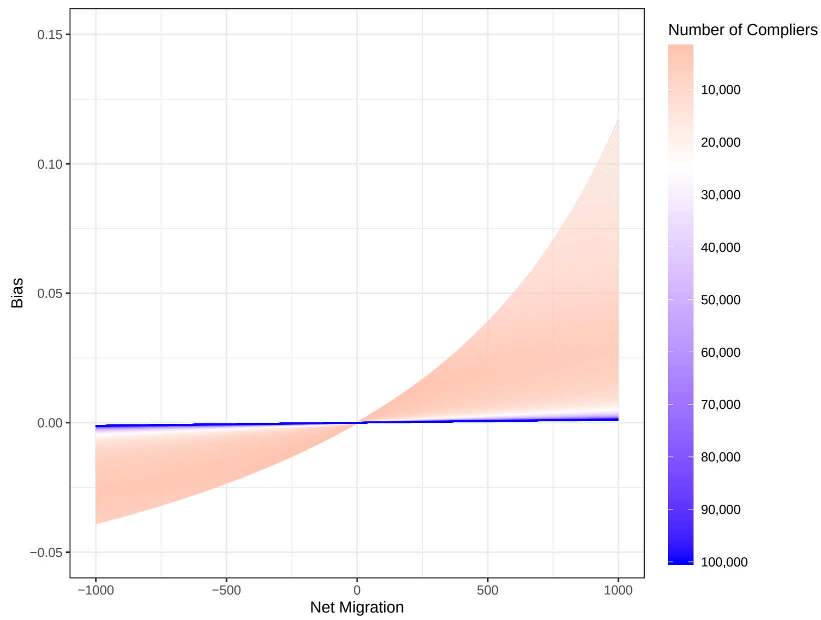

Figure A1 is a continuous surface rather than a set of points: the bias in the CACE as a function of net migration, one line per complier count, over five thousand counts. `maintained/output/figure_a1_bias.csv` holds that surface at the ten complier counts the legend labels, which is what a reader can read off the page.

---

9 Fixed-effects and random-effects meta-analysis

The published analysis pools state-level CACE estimates by fixed-effects inverse-variance weighting, which treats the true effect as identical across states and attributes all between-state variation to sampling error. The rewrite reports random-effects estimates alongside. This is an addition rather than a correction: every published fixed-effects number reproduces, and the random-effects figures sit beside them so a reader can see what the pooling assumption costs.

Table 7: Fixed-effects and random-effects meta-analytic CACE estimates for the Table 4 and Table 5 windows.

Pair type	Window	FE est.	FE SE	RE est.	RE SE
Mid-on-Mid	02-06	0.232	0.019	0.229	0.022
	06-10	0.119	0.009	0.141	0.021
	94-98	0.075	0.037	0.075	0.037
	98-02	0.213	0.026	0.256	0.052

	02-04	0.200	0.029	0.217	0.041
	06-08	0.166	0.018	0.200	0.030
Mid-on-Pres	10-12	0.111	0.019	0.110	0.025
	94-96	0.068	0.046	0.068	0.047
	98-00	0.226	0.036	0.337	0.128
	00-02	0.127	0.006	0.129	0.009
	04-06	0.109	0.003	0.140	0.016
Pres-on-Mid	08-10	0.090	0.002	0.097	0.008
	92-94	0.147	0.013	0.159	0.025
	96-98	0.138	0.011	0.171	0.061
	00-04	0.181	0.016	0.198	0.027
	04-08	0.117	0.007	0.122	0.012
Pres-on-Pres	08-12	0.117	0.005	0.122	0.010
	92-96	0.210	0.023	0.212	0.043
	96-00	0.189	0.022	0.189	0.022

Table 8: Between-state heterogeneity in the discontinuity CACEs.

Window	N states	$\hat{\tau}^2$	I^2
NA	NA	NA	NA
:——	———:	:—————	:——

Estimated between-state variance is non-zero in most windows and I^2 exceeds fifty per cent in several. The random-effects estimates are close to the fixed-effects ones, but their standard errors are wider, sometimes by a factor of two. For the 2008-on-2012 window the substantive conclusion is unchanged and the precision of the pooled estimate should be read more cautiously than the published standard error alone suggests.

---

10 Rewrite verification

`run_all.R` was run twice from clean sessions and the whole of `maintained/output/` came back byte-identical, figure PDFs included: `blank_pdf_timestamps()` overwrites the wall-clock `/CreationDate` and `/ModDate` that R's `pdf()` device stamps into every figure, which are otherwise the only bytes that change between runs. Nothing in the pipeline draws at random, so there is no seed to pin and no dispersion to report.

`original/` is verified twice per run: once at the top of `run_all.R`, before anything executes, and once at the end. The first pass is a precondition; the second is what demonstrates that no script wrote into the deposit. Both passes check every file against its served MD5 and its byte size, and both refuse a directory holding anything the manifest does not list. Three of the deposit's 24 published checksums (`SMG2009.RData`, `turnoutrates.RData` and `votemargins.RData`, all ingested by Dataverse as tabular data) match neither the original bytes the archive serves nor the derived `.tab` it generates. The local copies are verbatim in both cases, so the discrepancy is in the published metadata rather than in the deposit; `download_original.R` gates on the served checksum and prints the disagreement.

---

11 R environment

Item	Value
R version	4.6.0
Platform	aarch64-apple-darwin23
Date run	2026-08-10